

Jaxion: A JAX package for Fuzzy Dark Matter

Philip Mocz  ¹✉

¹ Center for Computational Astrophysics, Flatiron Institute, 162 5th Avenue, New York, NY 10010, USA

✉ Corresponding author

DOI: [10.21105/joss.09542](https://doi.org/10.21105/joss.09542)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Arfon Smith](#)  

Reviewers:

- [@shubhagrawal30](#)
- [@chaitanyaafle](#)

Submitted: 23 October 2025

Published: 29 July 2026

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

We introduce *jaxion*, a Python library built on JAX for 3D numerical simulations of fuzzy dark matter (FDM), gas, and stars. Spectral, particle-mesh, and finite volume solvers are combined to model the various physics components, which are coupled through gravity. The code is scalable to multiple GPUs. JAX's automatic differentiation enables the simulations to be used with optimization and inference workflows. *jaxion* provides a flexible framework for rapid prototyping at scale and integration of simulations with inverse problems or hybrid physics-ML modeling.

Statement of need

jaxion is designed to be an open-source release of previous research code algorithms that have been used to investigate several aspects of FDM ([Amin et al., 2022](#); [Amin & Mocz, 2019](#); [Church et al., 2019](#); [Davies & Mocz, 2020](#); [Dome et al., 2023, 2023](#); [Foote et al., 2023](#); [Lancaster et al., 2020](#); [Luu et al., 2024, 2025](#); [Mocz et al., 2017, 2018, 2019, 2020, 2023](#); [Mocz & Szasz, 2021](#); [Painter et al., 2024](#); [Alvaro Pozo et al., 2024](#); [A. Pozo et al., 2025](#)). This new release, written in JAX ([Bradbury et al., 2018](#)), has the added advantage of being differentiable and deployable on multiple GPUs.

Astrophysics research has long relied on sophisticated simulation codes. Established codes include: Athena++ ([Stone et al., 2020](#)), Arepo ([Springel, 2010](#)), FLASH ([Fryxell et al., 2000](#)), RAMSES ([Teyssier, 2002](#)), GAMER ([Schive et al., 2018](#)), PyUltraLight ([Edwards et al., 2018](#)). Such codes enable detailed studies of gas dynamics, star formation, cosmological structure formation, and galaxy evolution. These tools employ a combination of grid-based, particle-based, and spectral methods to solve the governing equations of hydrodynamics, gravity, and additional physics.

Despite their successes, classical astrophysics codes are limited in their ability to interface with modern machine learning (ML) frameworks and support automatic differentiation. As ML and AI techniques are becoming more integrated with scientific fields, e.g. for parameter inference, model discovery, and hybrid physics-ML modeling, there is a growing need for simulation frameworks that are flexible and differentiable. *jaxion* fills this gap, by leveraging automatic differentiation, hardware acceleration, and seamless integration with ML workflows. Other recent developments of differentiable astrophysics code for various applications ranging from hydrodynamical simulations to modeling gravitational waves include ([Horowitz & Lukic, 2025](#); [Lanzieri et al., 2022](#); [Wong et al., 2023](#))

jaxion is a differentiable simulation library specifically designed for studying FDM coupled to baryons (stars and gas). FDM is a plausible dark matter candidate, modeled as a quantum wave-like field. It exhibits unique phenomena such as solitonic cores and granular interference patterns on kiloparsec scales ([Hui et al., 2017](#)). *jaxion*, with built-in automatic differentiability, is aimed to open new avenues for scientific discovery through gradient-based parameter

inference, optimization, and hybrid physics-ML modeling.

Overview of functionality

jaxion solves the following equations:

Component	Governing Equations	Numerical Method
Fuzzy Dark Matter	Schrodinger-Poisson	Spectral
Gas	Compressible Euler (isothermal)	Finite Volume
Stars	Collisionless N-body	Particle-Mesh
Gravity	Poisson equation	Spectral

in a 3D periodic domain. It can solve equations in physical or comoving (cosmological) coordinates. Users can additionally optionally add an external potential. Features will continue to expand in future releases, including self-interactions, multiple axion fields, other fluid equations of state, and sink particles.

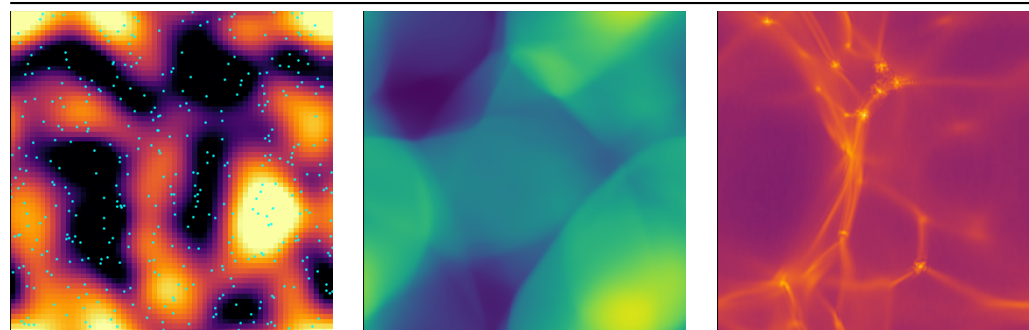
The code generates checkpoints (for restart and analysis) and images.

Documentation is found at: <https://jaxion.readthedocs.io/>

The Github Repository is at: <https://github.com/JaxionProject/jaxion>

Examples of simulation setups are found in the `examples/` directory, including inverse problems (optimization).

Below are snapshots from some of the examples:



Acknowledgements

We acknowledge discussions with researchers over the past decade which led to an initial implementation of FDM solvers, refinement and several publications over the years, conversations on differentiable simulations, and ultimately the development of this Python package. Thanks are due to: Mustafa A. Amin, Fernando Becerra, Gurtina Besla, Sownak Bose, Michael Boylan-Kolchin, Caroline Brickley, Pierre-Henri Chavanis, Benjamin V. Church, Jonathan Citrin, Elliot Yarnell Davies, Benedikt Diemer, Tibor Dome, Razieh Emami, Elisa G. M. Ferreira, Anastasia Fialkov, Hayden R. Foote, Dan Foreman-Mackey, Cara Giovanetti, Frank Graziani, Benjamin Hamm, Lars Hernquist, Ben Horowitz, Lam Hui, Tzanio Kolev, Lachlan Lancaster, Alvaro Pozo Larrocha, Mariangela Lisanti, Eric Ludwig, Hoang Nhan Luu, Tina Kahniashvili, Federico Marinacci, Simon May, Jens C. Niemeyer, Matthew Notis, Jerry Ostriker, Connor A. Painter, Jeff Parker, Rob Rieben, Victor H. Robles, Bodo Schwabe, Xuejian Shen, Martin Sparre, David Spergel, Volker Springel, Sauro Succi, Aaron Szasz, Hy Trac, Scott Tremaine, Mark Vogelsberger, Jesus Zavala.

References

- Amin, M. A., Jain, M., Karur, R., & Mocz, P. (2022). Small-scale structure in vector dark matter. *Journal of Cosmology and Astroparticle Physics*, 2022(8), 014. <https://doi.org/10.1088/1475-7516/2022/08/014>
- Amin, M. A., & Mocz, P. (2019). Formation, gravitational clustering, and interactions of nonrelativistic solitons in an expanding universe. *Physical Review D*, 100(6), 063507. <https://doi.org/10.1103/PhysRevD.100.063507>
- Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., & Zhang, Q. (2018). *JAX: Composable transformations of Python+NumPy programs* (Version 0.3.13). <http://github.com/google/jax>
- Church, B. V., Mocz, P., & Ostriker, J. P. (2019). Heating of Milky Way disc stars by dark matter fluctuations in cold dark matter and fuzzy dark matter paradigms. *Machine*, 485(2), 2861–2876. <https://doi.org/10.1093/mnras/stz534>
- Davies, E. Y., & Mocz, P. (2020). Fuzzy dark matter soliton cores around supermassive black holes. *Monthly Notices of the Royal Astronomical Society*, 492(4), 5721–5729. <https://doi.org/10.1093/mnras/staa202>
- Dome, T., Fialkov, A., Sartorio, N., & Mocz, P. (2023). Cosmic web dissection in fuzzy dark matter cosmologies. *Monthly Notices of the Royal Astronomical Society*, 525(1), 348–363. <https://doi.org/10.1093/mnras/stad2276>
- Edwards, F., Kendall, E., Hotchkiss, S., & Easther, R. (2018). PyUltraLight: a pseudo-spectral solver for ultralight dark matter dynamics. *Journal of Cosmology and Astroparticle Physics*, 2018(10), 027. <https://doi.org/10.1088/1475-7516/2018/10/027>
- Foote, H. R., Besla, G., Mocz, P., Garavito-Camargo, N., Lancaster, L., Sparre, M., Cunningham, E. C., Vogelsberger, M., Gómez, F. A., & Laporte, C. F. P. (2023). Structure, Kinematics, and Observability of the Large Magellanic Cloud's Dynamical Friction Wake in Cold versus Fuzzy Dark Matter. *Astrophysical Journal*, 954(2), 163. <https://doi.org/10.3847/1538-4357/ace533>
- Fryxell, B., Olson, K., Ricker, P., Timmes, F. X., Zingale, M., Lamb, D. Q., MacNeice, P., Rosner, R., Truran, J. W., & Tufo, H. (2000). FLASH: An Adaptive Mesh Hydrodynamics Code for Modeling Astrophysical Thermonuclear Flashes. *Astrophysical Journal Supplement Series*, 131(1), 273–334. <https://doi.org/10.1086/317361>
- Horowitz, B., & Lukic, Z. (2025). Differentiable Cosmological Hydrodynamics for Field-Level Inference and High Dimensional Parameter Constraints. *arXiv e-Prints*, arXiv:2502.02294. <https://doi.org/10.48550/arXiv.2502.02294>
- Hui, L., Ostriker, J. P., Tremaine, S., & Witten, E. (2017). Ultralight scalars as cosmological dark matter. *Physical Review D*, 95(4), 043541. <https://doi.org/10.1103/PhysRevD.95.043541>
- Lancaster, L., Giovanetti, C., Mocz, P., Kahn, Y., Lisanti, M., & Spergel, D. N. (2020). Dynamical friction in a Fuzzy Dark Matter universe. *Journal of Cosmology and Astroparticle Physics*, 2020(1), 001. <https://doi.org/10.1088/1475-7516/2020/01/001>
- Lanzieri, D., Lanusse, F., & Starck, J.-L. (2022). Hybrid Physical-Neural ODEs for Fast N-body Simulations. *Machine Learning for Astrophysics*, 60. <https://doi.org/10.48550/arXiv.2207.05509>
- Luu, H. N., Mocz, P., Vogelsberger, M., May, S., Borrow, J., Tye, S.-H. H., & Broadhurst, T. (2024). Nested solitons in two-field fuzzy dark matter. *Monthly Notices of the Royal Astronomical Society*, 527(2), 4162–4172. <https://doi.org/10.1093/mnras/stad3482>

- Luu, H. N., Mocz, P., Vogelsberger, M., Pozo, A., Broadhurst, T., Tye, S.-H. H., Liu, T., Fung, L. W. H., Smoot, G. F., Emami, R., & Hernquist, L. (2025). Diverse dark matter haloes in two-field fuzzy dark matter. *Phys. Rev. D*, 111(12), L121302. <https://doi.org/10.1103/w9j1-k7b3>
- Mocz, P., Fialkov, A., Vogelsberger, M., Becerra, F., Amin, M. A., Bose, S., Boylan-Kolchin, M., Chavanis, P.-H., Hernquist, L., Lancaster, L., Marinacci, F., Robles, V. H., & Zavala, J. (2019). First Star-Forming Structures in Fuzzy Cosmic Filaments. *Physical Review Letters*, 123(14), 141301. <https://doi.org/10.1103/PhysRevLett.123.141301>
- Mocz, P., Fialkov, A., Vogelsberger, M., Becerra, F., Shen, X., Robles, V. H., Amin, M. A., Zavala, J., Boylan-Kolchin, M., Bose, S., Marinacci, F., Chavanis, P.-H., Lancaster, L., & Hernquist, L. (2020). Galaxy formation with BECDM - II. Cosmic filaments and first galaxies. *Monthly Notices of the Royal Astronomical Society*, 494(2), 2027–2044. <https://doi.org/10.1093/mnras/staa738>
- Mocz, P., Fialkov, A., Vogelsberger, M., Boylan-Kolchin, M., Chavanis, P.-H., Amin, M. A., Bose, S., Dome, T., Hernquist, L., Lancaster, L., Notis, M., Painter, C., Robles, V. H., & Zavala, J. (2023). Cosmological structure formation and soliton phase transition in fuzzy dark matter with axion self-interactions. *Monthly Notices of the Royal Astronomical Society*, 521(2), 2608–2615. <https://doi.org/10.1093/mnras/stad694>
- Mocz, P., Lancaster, L., Fialkov, A., Becerra, F., & Chavanis, P.-H. (2018). Schrödinger-Poisson-Vlasov-Poisson correspondence. *Physical Review D*, 97(8), 083519. <https://doi.org/10.1103/PhysRevD.97.083519>
- Mocz, P., & Szasz, A. (2021). Toward Cosmological Simulations of Dark Matter on Quantum Computers. *Astrophysical Journal*, 910(1), 29. <https://doi.org/10.3847/1538-4357/abe6ac>
- Mocz, P., Vogelsberger, M., Robles, V. H., Zavala, J., Boylan-Kolchin, M., Fialkov, A., & Hernquist, L. (2017). Galaxy formation with BECDM - I. Turbulence and relaxation of idealized haloes. *Monthly Notices of the Royal Astronomical Society*, 471(4), 4559–4570. <https://doi.org/10.1093/mnras/stx1887>
- Painter, C. A., Boylan-Kolchin, M., Mocz, P., & Vogelsberger, M. (2024). An attractive model: simulating fuzzy dark matter with attractive self-interactions. *Monthly Notices of the Royal Astronomical Society*, 533(2), 2454–2472. <https://doi.org/10.1093/mnras/stae1912>
- Pozo, Alvaro, Broadhurst, T., Smoot, G. F., Chiueh, T., Luu, H. N., Vogelsberger, M., & Mocz, P. (2024). Dwarf galaxies united by dark bosons. *Phys. Rev. D*, 109(8), 083532. <https://doi.org/10.1103/PhysRevD.109.083532>
- Pozo, A., Emami, R., Mocz, P., Broadhurst, T., Hernquist, L., Vogelsberger, M., Smith, R., Tremblay, G., Narayan, R., Steiner, J., Grindlay, J., & Smoot, G. (2025). Galaxy formation with wave/fuzzy dark matter: The core-halo structure and the solitonic imprint. *A&A*, 699, A308. <https://doi.org/10.1051/0004-6361/202450443>
- Schive, H.-Y., ZuHone, J. A., Goldbaum, N. J., Turk, M. J., Gaspari, M., & Cheng, C.-Y. (2018). GAMER-2: a GPU-accelerated adaptive mesh refinement code - accuracy, performance, and scalability. *Monthly Notices of the Royal Astronomical Society*, 481(4), 4815–4840. <https://doi.org/10.1093/mnras/sty2586>
- Springel, V. (2010). E pur si muove: Galilean-invariant cosmological hydrodynamical simulations on a moving mesh. *Monthly Notices of the Royal Astronomical Society*, 401(2), 791–851. <https://doi.org/10.1111/j.1365-2966.2009.15715.x>
- Stone, J. M., Tomida, K., White, C. J., & Felker, K. G. (2020). The Athena++ Adaptive Mesh Refinement Framework: Design and Magnetohydrodynamic Solvers. *Astrophysical Journal Supplement Series*, 249(1), 4. <https://doi.org/10.3847/1538-4365/ab929b>
- Teyssier, R. (2002). Cosmological hydrodynamics with adaptive mesh refinement. A new

high resolution code called RAMSES. *Astronomy and Astrophysics*, 385, 337–364. <https://doi.org/10.1051/0004-6361:20011817>

Wong, K. W. K., Isi, M., & Edwards, T. D. P. (2023). Fast Gravitational-wave Parameter Estimation without Compromises. *Astrophysical Journal*, 958(2), 129. <https://doi.org/10.3847/1538-4357/acf5cd>